

# Rajat Mehndiratta

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## Skills

Languages: Python, OCaml, Java, Rust, Go, Swift, TypeScript  
Tools: git, Google Sheets, Mongo, React, SwiftUI,  $\text{\LaTeX}$

## Experience

### Software Engineer

April 2023 - October 2024

Jane Street Capital > Options Desk > Proprietary Trading System Development

Collaborated with traders in fast-paced environment to provide liquidity across multiple markets.  
+ Collaborated on greenfield expansion into new index and equity options.  
+ Led infrastructure migration and quality of life efforts with impact across multiple trading systems.  
+ Owned several proprietary functionalities and built new model training tooling.

### Software Engineer

August 2021 - February 2023

Plaid > Internal Platform > Core Services > Abstractions

Prioritized correctness, reliability, and velocity in data modeling and storage for core systems in Go.  
+ Owned project to resolve scaling limit, extending company storage runway from 2022Q1 to late 2020's.  
+ Built verification to help detect and manage 100MM's of obsolete user connections, mitigating regulatory and security risk while increasing internal observability around sensitive user data.  
+ Led component efforts for infrastructure cost reduction (\$45k/mo) and incident response speedup.

### Software Engineer

June 2019 - July 2021

Google > Tech Infra > Unified Fulfillment Optimization > Fleet Transformation

Enhanced Java backend for mixed-integer programming solver microservice to plan datacenters, focused on reducing lead time variance and increasing granularity and touchlessness of planning.  
+ Launched recurring automated jobs to detect and re-plan no-longer-viable hardware deployments.  
+ Designed and delivered API to plan with counterfactuals, increasing on-time fulfillment and reducing need for manual monitoring and intervention in hardware planning and deployment.

### Software Development Engineer Intern

May - August 2018

Amazon > Supply Chain Optimization Technologies > Topline Forecasting

Investigated and demonstrated serverless solutions to reduce forecasting compute costs by 10-100x.  
+ Designed and implemented Python workflow orchestrator to operate serverless forecasting on AWS.

### Software Development Engineer Intern

May - August 2017

Amazon > Worldwide Operations > Robotics

Validated deep reinforcement learning for visual navigation in sidewalk delivery (Amazon Scout).  
+ Drove effort to generate training data from realistic commercial virtual 3D environments.

## Education

### Carnegie Mellon University

May 2019

Bachelor of Science, Electrical and Computer Engineering

Coursework: Advanced Mobile Robot Development (16-865), Advanced Digital Signal Processing (18-792), Introduction to Machine Learning (10-601), Rapid Prototyping of Computer Systems (18-540), Introduction to Computer Architecture (18-447), Natural Language Processing (11-411), Neural Technology: Sensing & Stimulation (18-412), Logic Design and Verification (18-341)  
Involvements: hackathons (competitor, mentor, organizer, and sponsor), Mock Trial (Captain), SDC Buggy (carbon-fiber gravity racing; Mechanic), End The Rain (umbrella dispensers; Co-Founder, Tech Lead)

## Projects

### SCOT-T Lunar Rover (16-865 / CMU+Astrobotic Lunar X Prize Entry)

January - May 2016

Engineer on Communications, Hardware, and UI Teams

Worked on development and testing of UDP-based communications for novel 4.5kg Cube Rover standard.

### FifthSense (PennApps Fall 2015 Grand Prize)

September 6-8, 2015

Hackathon Competitor

Built full-duplex handheld design to allow blind users to access smartphones while moving around.